BNKE: Kernel-Sensitive Shot Allocation for Projected Quantum Kernels

Under Finite Measurement Budgets

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ICTAI 2026

Why shot allocation matters

A shot is one repetition of a quantum measurement. More shots mean more precise estimates.

18

FEATURES PER IMAGE

X, Y and Z measurements on each of six qubits

500K

TRAINING SHOTS

A single fixed budget shared by all measured features

≈5

SHOTS PER ENTRY

Uniform measurement at 5,000 samples (before rounding)

The question: can the same 500,000 shots be placed more intelligently?

Uniform: equal attention to every feature • BNKE: focus on influential features

What we contribute

A constrained measurement problem, a principled rule, and an empirical ablation

Four contributions — each within the same controlled PQK setting

01 Allocation framework

Convert per-entry importance into a fixed-budget shot schedule.

02 Surrogate theory

Square-root allocation is optimal for a specified weighted estimation surrogate.

03 Empirical evidence

Sensitivity wins over uniform across all five MedMNIST benchmarks.

04 Numerical stability

Ridge regularization makes exact-versus-noisy kernel comparisons reliable.

Important: surrogate optimality does not imply an accuracy guarantee.

Same kernel model; only shot allocation changes

Six PCA inputs → shallow circuit → Pauli measurements → RBF kernel → SVM



Only the training-side shot allocation changes between methods.

Allocate shots in proportion to √importance

The budget stays fixed; more influential entries receive more measurements.

$$S^*_{ik} = B \times \sqrt{w_{ik}} \,/\, (\text{sum of } \sqrt{w} \text{ over all entries})$$

- B is the fixed training budget; w_{ik} is a nonnegative importance score.
- Integer allocation enforces at least one shot per entry.
- The rule minimizes $\Sigma\, w_{ik} / S_{ik}$ under a continuous budget constraint.

Five ways to decide where shots go

Only the importance signal changes across the five allocation policies.

BNKE-U	Uniform shots
BNKE-V	High measurement variance
BNKE-B	Near SVM boundary
BNKE-S	Large kernel sensitivity
BNKE-SV	Sensitivity × variance

Kernel sensitivity targets impact—not just noise

Variance asks how uncertain a feature is; sensitivity asks how much it matters.

ORACLE STUDY

$$w_{ik}(\text{sensitivity}) = \Sigma_j K_{ij}^2 (f_{ik} - f_{jk})^2$$

Variance

How noisy is an estimate?

Sensitivity

How much does that coordinate influence kernel similarity?

Sensitivity is derived by first-order perturbation analysis; it is not an exact accuracy gradient.

Controlled experiment across five medical datasets

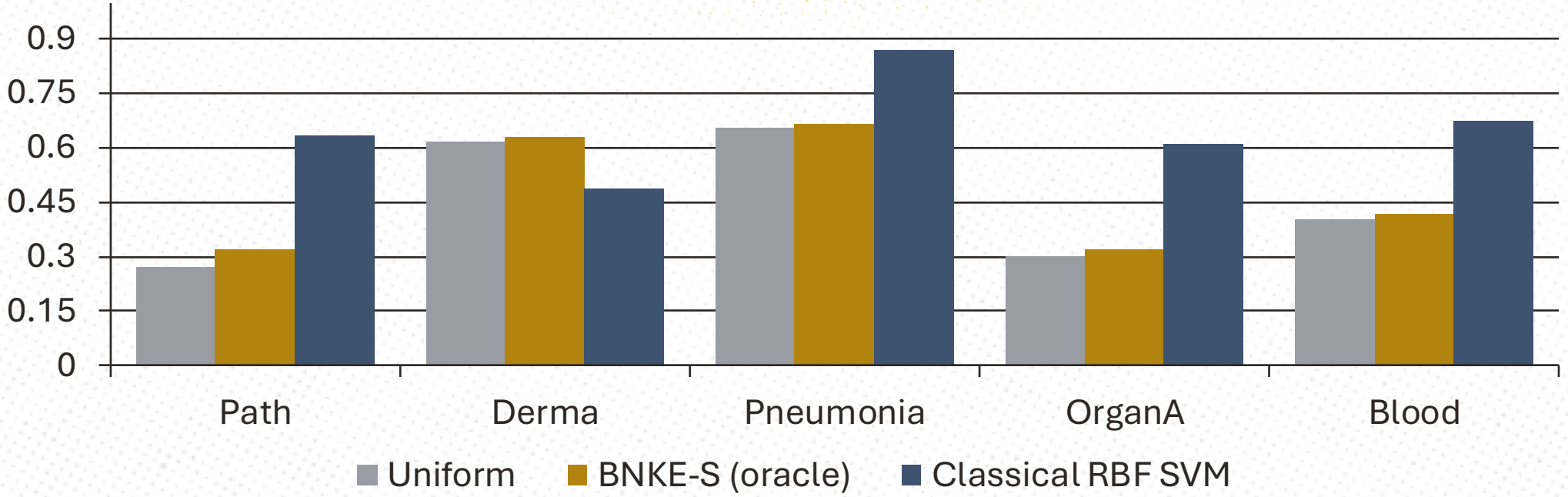
All methods share the split, circuit, kernel and SVM setup.

- Five MedMNIST v2 datasets; up to 5,000 training and 1,000 test images.
- 6-qubit, depth-1 circuit; 18 features; RBF bandwidth $\gamma = 0.5$.
- Main comparison: $B = 500{,}000$ training shots; 30 paired noise runs.
- Fixed indices, exact features, SVM settings, and shared test noise.
- Gaussian shot-noise model checked against exact Bernoulli sampling.

Sensitivity improves accuracy on every benchmark

Five MedMNIST datasets • 500,000 training shots • 30 paired runs

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Oracle BNKE-S − Uniform: +0.046 | +0.013 | +0.015 | +0.022 | +0.019

Paired gains survive Holm correction ($p \leq 5.8 \times 10^{-4}$). Classical baseline uses PCA features.

Only sensitivity-based weights consistently help

Variance-only and decision-boundary allocation offer no reliable gain.

Variance-only

No consistent gain over uniform; worse on DermaMNIST.

Boundary-only

No consistent gain; shares one importance value across all observables of a sample.

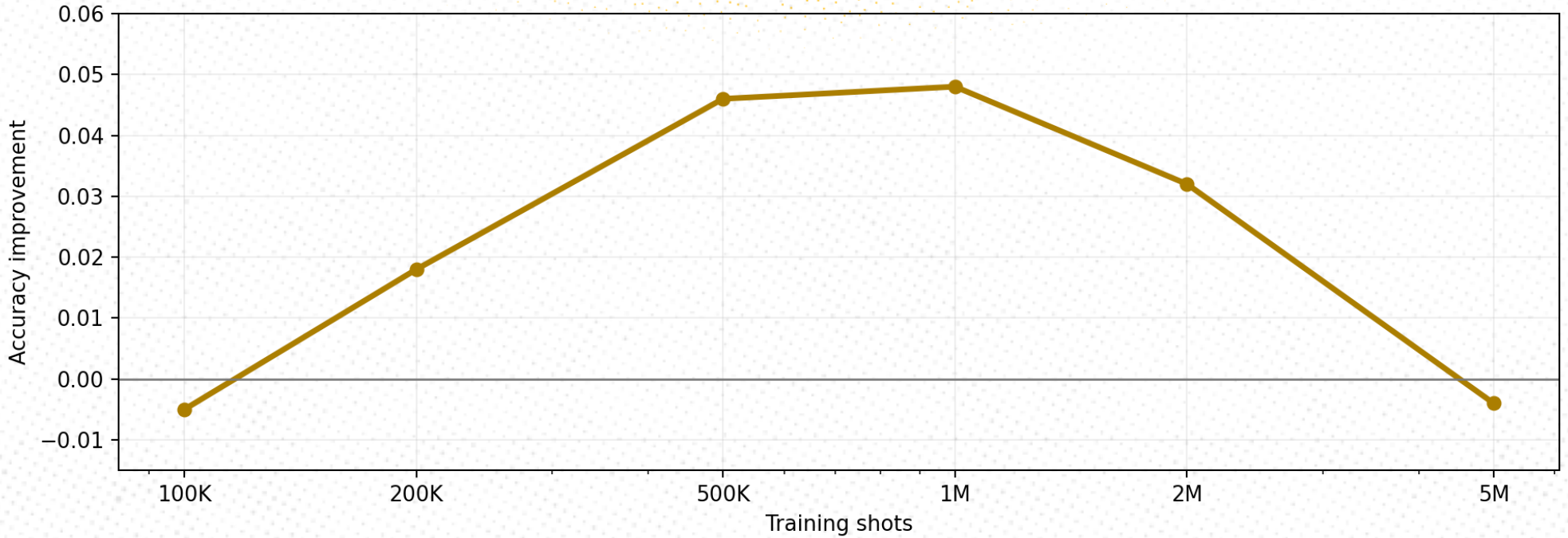
Sensitivity × variance

Improves versus uniform, but not significantly better than sensitivity alone.

BNKE-S is the simpler choice. Across datasets, it recovers only 7–27% of the exact-PQK finite-shot accuracy loss.

The advantage is greatest at intermediate budgets

PathMNIST: the one-shot floor limits low budgets; abundant shots reduce the need for reallocation.



Low budget: one-shot floor binds • High budget: uniform estimates become adequate

The result survives two important checks

Alternative data splits and regularized kernel conditioning.

Split robustness

Five alternative splits per dataset × 10 noise runs. BNKE-S has a positive mean paired difference in all 25 dataset–split combinations.

Kernel conditioning

Ridge stabilizes poorly conditioned kernels. Without ridge, exact kernels can misleadingly appear worse than noisy ones.

Checks support the controlled finding, not generalization to unseen circuits or hardware.

A theorem about estimation—not classification

Separate what is guaranteed mathematically from what is observed experimentally.

What we prove

For fixed nonnegative weights, square-root allocation minimizes the weighted surrogate in the continuous setting; its surrogate loss is no worse than uniform.

What we observe

Under the specified simulated PQK pipeline, oracle sensitivity allocation raises test accuracy relative to uniform on five benchmarks.

We do NOT prove improved classification accuracy or quantum advantage.

The main open problem is measuring sensitivity

Oracle scores come from exact simulated features, not a finite-shot hardware pilot.

- Oracle sensitivity needs exact features that hardware cannot supply for free.
- A practical system needs a pilot measurement budget to estimate sensitivity.
- All reported results use simulation, six qubits, one shallow circuit, and one RBF kernel.
- Next: jointly optimize pilot and allocation budgets; test deeper circuits and real devices.

What should you remember?

Measurement placement—not only shot count—can matter.

01

A fixed shot budget can be allocated better than uniformly.

02

Kernel sensitivity is a stronger guide here than variance or classifier margins.

03

The measured benefit is real in simulation—but the oracle-to-pilot gap remains open.

Thank you — I welcome questions.

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ACKNOWLEDGMENT

We thank the anonymous reviewers for their feedback.

This research was in part supported by the US NSF under Grants: SHF-2210744, CSR-2549533, CNS-2244450, and AMPS-2229073.